Part 2: Tooth Growth by Supplement and Dose

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Data and exploratory analysis

`ToothGrowth` records tooth length (`len`) of 60 guinea pigs given vitamin C at three doses (0.5, 1, 2 mg/day) by orange juice (OJ) or ascorbic acid (VC); 10 animals per supp-dose cell.

Supp	Dose	n	Mean len	SD
OJ	0.5	10	13.23	4.46
VC	0.5	10	7.98	2.75
OJ	1	10	22.70	3.91
VC	1	10	16.77	2.52
OJ	2	10	26.06	2.66
VC	2	10	26.14	4.80

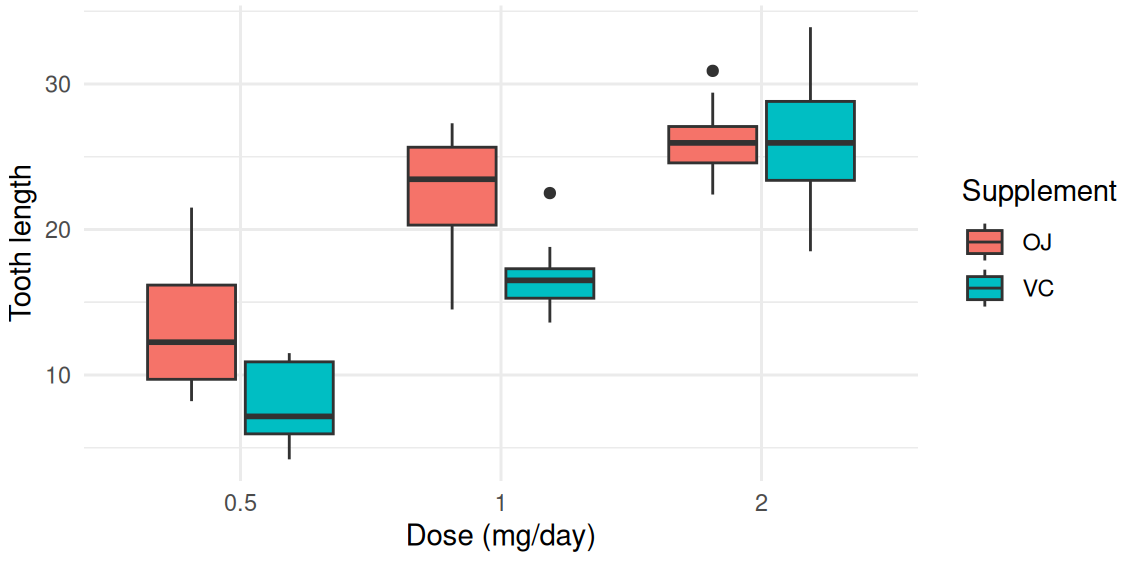


Figure 1: Tooth length by dose and supplement.

Length rises clearly with dose for both supplements. OJ appears higher at 0.5 and 1 mg, while at 2 mg the two look similar.

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Comparisons

I use two-sample t confidence intervals / tests (unequal variances, Welch, the default of `t.test`), 95% level. Supplement comparisons are unpaired (different animals).

Comparison	Mean diff	CI low	CI high	p-value
OJ vs VC, all doses	3.700	-0.171	7.571	0.061
OJ vs VC, dose 0.5	5.250	1.719	8.781	0.006
OJ vs VC, dose 1	5.930	2.802	9.058	0.001
OJ vs VC, dose 2	-0.080	-3.798	3.638	0.964
Dose 1 vs 0.5 (all supp)	9.130	6.276	11.984	<0.001
Dose 2 vs 1 (all supp)	6.365	3.734	8.996	<0.001
Dose 2 vs 0.5 (all supp)	15.495	12.834	18.156	<0.001

Conclusions

- **Dose:** each increase in dose raises mean tooth length significantly; all three dose comparisons have CIs entirely above 0 and $p < 0.001$. Doubling from 0.5 to 2 mg adds about 15 units.
- **Supplement, overall:** OJ is higher by about 3.7 units, but the CI (about -0.17 to 7.57) includes 0 ($p \approx 0.06$), so pooled across doses we cannot conclude a difference at the 5% level.
- **Supplement within dose:** OJ gives significantly longer teeth at 0.5 mg ($p \approx 0.006$) and 1 mg ($p \approx 0.001$); at 2 mg there is no evidence of a difference (CI includes 0, $p \approx 0.96$). So the OJ advantage appears at lower doses and vanishes at the highest.

Assumptions. Animals are independent and randomly assigned to supplement/dose; lengths are approximately normal within groups (or samples large enough for the t approximation, a limitation with n = 10 per cell); variances may differ (hence Welch). Comparisons are not corrected for multiple testing (7 tests), so results should be read cautiously. The dose effects and the OJ effects at 0.5 mg ($p \approx 0.006$) and 1 mg survive a Bonferroni threshold of $0.05/7 \approx 0.007$, though 0.5 mg only just.

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Appendix: code

```
data(ToothGrowth); tg <- ToothGrowth; tg$dose <- factor(tg$dose)
t.test(len ~ supp, data = tg)                                  # pooled over dos
e
t.test(len ~ supp, data = subset(tg, dose == 0.5))             # per dose
t.test(tg$len[tg$dose==1], tg$len[tg$dose==0.5])               # dose contrasts
```

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